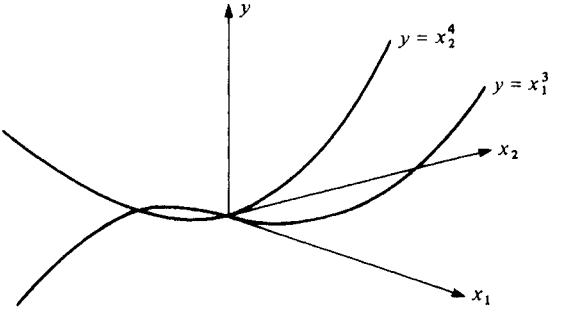
(iii) Since $\det f''(0,0) = 0$, the method is useless for classifying the stationary point $(0, 0)$. Observe, however, that

$$y = f(0, x_2) = x_2^4$$
$$y = f(x_1, 0) = x_1^3$$

and so the function certainly does not have a local maximum or minimimum at $(0, 0)$.



19.46 Example Consider the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1$. The stationary points are found by solving the equations

$$\left.\begin{aligned} x_2 + x_3 &= 0 \\ x_1 + x_3 &= 0 \\ x_2 + x_1 &= 0 \end{aligned}\right\}$$

simultaneously. The only stationary point is therefore $(0, 0, 0)$. The second derivative at this point is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

which has principal minors $2, -1$ and 0. The matrix is neither positive definite nor negative definite because one of the principal minors is zero. But $f''(0, 0, 0) \neq 0$ and so we have a saddle point at $(0, 0, 0)$.

19.47 Exercise

(1) A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f(x) = (Ax + a)^{\mathrm{T}}(Bx + b)$$

where A and B are $m \times n$ matrices and x, a and b are $m \times 1$ column

vectors. Find a system of linear equations satisfied by the local maxima and minima of f. [Hint: example 19.30.]

(2) Show that that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1 e^{-x_2}$$

is twice differentiable at every point and determine $f''(x_1, x_2)$.

(3) Show that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^3 x_2}{x_1^2 + x_2^2} & (x_1, x_2) \neq (0,0) \\ 0 & (x_1, x_2) = (0,0) \end{cases}$$

has a continuous first derivative $f'\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(0,0)$ and that all its second order partial derivatives exist at $(0,0)$. Prove that

$$f_{12}(0,0) \neq f_{21}(0,0)$$

and explain why this does not contradict proposition 19.33.

(4) Classify the stationary points of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1 - x_1 x_2$$

using the second derivative. [Hint: example 19.10.]

(5) Show that there exists no value of $\xi \in (0, 1)$ such that

$$f(1) - f(0) = f'(\xi)(1 - 0)$$

in the case of the function $f\colon \mathbb{R}^1 \to \mathbb{R}^2$ defined by

$$f(t) = \begin{pmatrix} t^2 \\ t^3 \end{pmatrix}.$$

(See §19.34.)

(6) Suppose that λ is an eigenvalue of an $n \times n$ symmetric matrix M with real entries. Using the results quoted in this chapter, prove that

$$\lambda^2 \leqslant \sum_{i=1}^{n} \sum_{j=1}^{n} m_{ij}^2.$$

19.48† Differentials

The derivative $f'(x)$ or $Df(x)$ of a vector function f at a point x was defined to be a matrix. Recall that f is differentiable at x if and only if there exists an affine function g which is tangent to f at x. The affine function g is given by

$$g(x + h) = f(x) + f'(x)h.$$